Collaborative Multi-Agent Scripts Generation for Enhancing Imperfect-Information Reasoning in Murder Mystery Games

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Abstract

Vision-language models (VLMs) have shown impressive capabilities in perceptual tasks, yet they degrade in complex multi-hop reasoning under multi-player game settings with imperfect and deceptive information. In this paper, we pick up a representative multi-player task, Murder Mystery Games, which require to infer hidden truths based on partial clues provided by the roles of different intentions. To address this challenge, we propose a collaborative multi-agent framework for evaluating and synthesizing high-quality, role-driven multi-player game scripts, enabling fine-grained interaction patterns tailored to character identities (i.e., murderer vs. innocent). Our system generates rich multimodal contexts—including character backstories, visual/textual clues, and multi-hop reasoning chains—through coordinated agent interactions. We design a two-stage agent-monitored training strategy to enhance the reasoning ability of VLM: (1) Chain-of-Thought based fine-tuning on curated and synthetic datasets that model uncertainty and deception; (2) GRPO-based Reinforcement Learning with agent-monitored reward shaping, encouraging the model to develop character-specific reasoning behaviors and effective multi-modal multi-hop inference. Extensive experiments demonstrate that our method significantly boosts the performance of VLM in narrative reasoning, hidden fact extraction, and deception-resilient understanding. Our contributions offer a scalable solution for training and evaluating VLMs under uncertain, adversarial, and socially complex conditions, laying the groundwork for future benchmarks in multimodal multi-hop reasoning under imperfect information.

1 Introduction

Vision-language models (VLMs) have demonstrated impressive capabilities in foundational perceptual tasks such as image captioning and visual question answering (VQA), as well as in more complex reasoning tasks through chain-of-thought (CoT) prompting, leveraging their ability to align and integrate information across visual and linguistic modalities (OpenAI, 2023; Google, 2023; Li et al., 2025b; Wang et al., 2025). However, tasks that demand sophisticated reasoning particularly those involving multi-hop inference, imperfect or deceptive information, and dynamic social interactions—remain challenging (Yang et al., 2018; Chen et al., 2024a,b). Recent efforts have begun to explore more expressive and controllable reasoning paradigms, to overcome bottlenecks in representation and reasoning capacity (Dong et al., 2025, 2026; Jiang et al., 2026). Advancing VLMs in such settings necessitates evaluation and training environments that require not only perception and knowledge, but also deeper reasoning and adaptability under imperfect information.

In real life, many practical tasks involve a multi-player game-theoretic process using imperfect information. For example, in judicial proceedings, judges, prosecutors, defense lawyers, witnesses, and juries engage in multiple rounds of social interaction from their respective perspectives, and make multi-hop inferences and decisions based on incomplete information, ultimately attempting to finish their own task.

To study such imperfect-information multi-player reasoning in vision-language models, we adopt Murder Mystery as a representative test environment. Murder Mystery is a social deduction role-playing game in which players assume predefined identities and collaboratively infer the hidden murderer, making it a typical yet tractable setting for modeling multi-agent interaction and reasoning under uncertainty. With access to public and private textual and visual clues, players engage in structured dialogue to reason about motives and inconsistencies, and infer the murderer amid adversarial deception. The game proceeds through four

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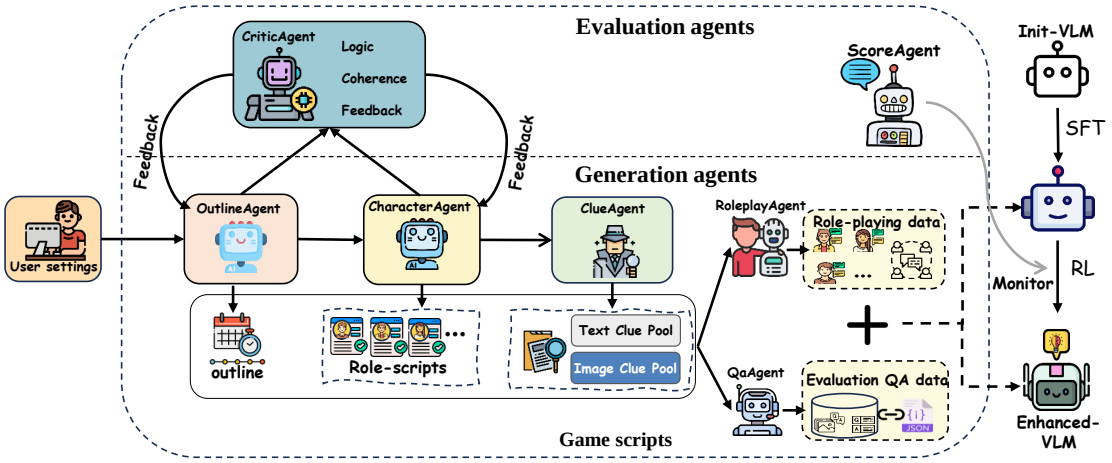

Figure 1: Overview of the proposed framework. It employs evaluation agents and generation agents to collaboratively generate logically coherent game scripts and instructs a pretrained VLM via a two-stage training strategy under agent monitoring to enhance the target model's reasoning capability under imperfect information.

key phases:

1. **Role Setup and Clue Absorption:** Players receive the rules and character backgrounds, followed by textual and visual clues. They then provide in-character self-introductions as the begin.

2. **Interactive Discussion:** Players engage in question-and-answer interaction with each other based on their clues and suspicions, emphasizing social inference, inconsistency detection, credibility assessment, and information selection.

3. **Hypothesis Generation:** Integrating accumulated clues and dialogue, players generate reasoning chains to infer motives and methods. This phase requires multi-hop multimodal reasoning across narrative and visual content.

4. **Final Decision:** Each player makes a final judgment regarding the murderer's identity.

This setting embodies key challenges such as imperfect information, inconsistency detection, and strategic social interaction, making it a suitable testbed for evaluating multi-hop multimodal reasoning in vision-language models. In addition, the task probes models' abilities in long-form narrative understanding, multimodal evidence integration, and the synthesis of textual and visual information through multi-step inference.

Although Murder Mystery is a representative task for modeling multi-player game process, there is still a lack of large-scale datasets for fine-tuning and evaluating models. Large-scale production of high-quality murder mystery scripts is expensive and impractical. To address this challenge, we design a multi-agent simulation framework where powerful LLMs (e.g., Gemini 2.5Pro (Gemini Team, 2025)) act as autonomous agents to collaboratively synthesize diverse Murder Mystery game scripts, producing challenging questions-answering pairs and multi-player interactive dialogue as training datasets(Ma et al., 2026; Zhang et al., 2025c). To enable the fine-tuning of VLMs on complex and adversarial examples, we build a new paradigm to automatically generate reasoning chains based on incomplete information. Lastly, we adopt a scalable two-stage training pipeline to learn VLMs(Han et al., 2026), via combining high-quality, auto-generated cases with curated training data.

Our main contributions are summarized as follows:

- **Multi-Agent Script Synthesis Framework:** We propose a scalable multi-agent framework to automatically generate diverse, high-quality multi-player game scripts. This framework simulates realistic character roles, player interactions and multimodal clues.

- **Training Data Construction and Learning under Imperfect Information:** We develop a novel paradigm for generating reasoning chains under imperfect information, enhancing model learning via a two-stage agent-monitored strategy.

- **Performance Enhancement under Imper-**

fect Information: Our method demonstrates consistent performance gains in reasoning and role-playing for vision-language models at both the 3B and 7B scales (e.g., Qwen2.5-VL-3B-Instruction and its 7B counterpart) in Murder Mystery scenarios involving imperfect and deceptive information.

2 Related Work

Social Reasoning Games as VLM Evaluation Platforms Social reasoning games, such as Murder Mystery and Werewolf, have become prominent platforms for evaluating the reasoning capabilities of VLMs in settings characterized by imperfect information, multi-agent interactions, and deception. These games provide a robust framework for assessing cognitive resilience in complex, multimodal scenarios (Zhu et al., 2025; Wu et al., 2024). WhodunitBench, offers 50 murder myster scripts with both multiple-choice and open-ended questions to facilitate multi-agent reasoning assessment (Xie et al., 2024). Frameworks such as MultiMind extend the evaluation to non-verbal modalities, incorporating facial expressions and intonation (Zhang et al., 2025e). The SocialMaze benchmark focuses on VLM reasoning in static social contexts, explicitly excluding deceptive elements (Xu et al., 2025b). Other frameworks, including BALROG, KORGym, and VS-Bench, assess multimodal reasoning in dynamic game environments but do not explicitly target social interaction capabilities (Paglieri et al., 2024; Shi et al., 2025; Xu et al., 2025a).

Multi-Agent Synthetic Data The scarcity of high-quality multimodal training data remains a significant bottleneck for VLM development. Synthetic data generation, particularly through multi-agent systems, has emerged as a scalable solution that enhances dataset diversity and reasoning complexity while reducing reliance on manual annotation(Ma et al., 2026; Zhang et al., 2025c). AgentInstruct utilizes a hierarchical multi-agent workflow to automatically produce synthetic instruction-response data with minimal human involvement (Mitra et al., 2024). Similarly, MATRIX simulates multi-agent social scenarios to generate data for alignment and instruction tuning (Tang et al., 2025). AudioGenie uses a dual-team multi-agent framework consisting of a "generation team" and a "supervision team" to generate diverse audio from multimodal inputs(Rong et al., 2025). Frameworks such as GenArtist and LayerCraft operate on similar principles. GenArtist decomposes complex text prompts into sub-tasks using a VLM-based agent, constructs detailed planning trees, and leverages external tools (e.g., SDXL, DALL-E 3) for image generation and editing. Iterative verification and self-correction further enhance output fidelity (Wang et al., 2024; Zhang et al., 2025b,d). Recent works on composed image retrieval further highlight the importance of modeling fine-grained modification signals and compositional semantics for generating high-quality multimodal data (Li et al., 2026; Chen et al., 2025, 2026; Zhang et al., 2026a; Qiu et al., 2026).

Training Pipelines for Reasoning-Enhanced VLMs Recent research frequently adopts a supervised fine-tuning (SFT) followed by reinforcement learning (RL) pipeline to enhance VLM reasoning. Both Reason-RFT (Tan et al., 2025) and SRPO (Zhang et al., 2025a) employ this two-stage approach: SFT is used to instill structured chain-of-thought reasoning, while RL further optimizes reasoning quality and generalization(Dong et al., 2025, 2026; Jiang et al., 2026). In a curriculum-based paradigm, Infi-MMR (Liu et al., 2025) progressively transitions from textual to multimodal and caption-free reasoning using sequential RL, achieving strong results on multimodal math benchmarks. VILASR introduces a "drawing-to-reason" paradigm, utilizing simple visual operations (e.g., auxiliary lines) to articulate spatial relationships, and employs a three-stage training process—synthetic data pre-training, reflective rejection sampling, and RL—to improve self-correction and generalization (Wu et al., 2025). Recent efforts also explore retrieval-augmented and experience-driven learning paradigms to improve long-horizon reasoning and interaction efficiency. For instance, ExpSeek proposes a self-triggered experience seeking mechanism for web agents, enabling adaptive data acquisition and policy refinement during training (Zhang et al., 2026b). Meanwhile, self-supervised and self-improving training paradigms further push toward unified multimodal learning without heavy human annotation (Han et al., 2026).

3 Method

This section first describes our multi-agent framework, and then takes the Murder Mystery Games as the application scenario to depict the process of

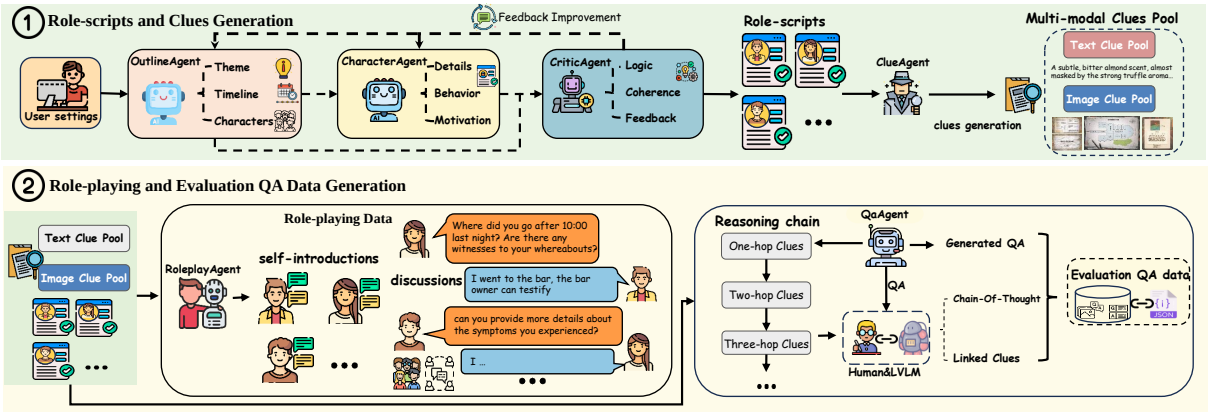

Figure 2: The details of game scripts generated via our multi-agent framework.

applying our framework.

3.1 Overview of Proposed Multi-Agent Framework

The proposed collaborative multi-agent framework aims to leverage collaborative agents to generate high-quality training data and instruct a pretrained VLM to enhance its reasoning under imperfect information in game-theoretic tasks.Our multi-agent framework includes two types of agents, i.e., **generation agents** and **evaluation agents**. While generation agents simulate realistic, interactive game processes to generate game-script data, evaluation agents focus on assessing the quality of these generated outputs and offering constructive feedback for improvement.

As shown in Figure 1, the framework includes generation agents such as the OutlineAgent, which produces story outlines with background and role summaries; the CharacterAgent, which creates detailed role scripts; and the ClueAgent, which generates multimodal clues that convey key environmental information. Building upon these elements, the RoleplayAgent produces role-playing data for specific scenarios, while the QaAgent constructs question–answer pairs to assess and strengthen the model's reasoning ability. To ensure quality, the CriticAgent evaluates the generated scripts for logical coherence and behavioral consistency. During training, the ScoreAgent assesses the model's role-specific behaviors, measuring how well its interactions align with the designated roles, and uses this feedback to facilitate model improvement. All agents interact through shared game scripts, working collaboratively to enhance imperfect-information reasoning, and the framework remains extensible for diverse game-theoretic tasks by adapting or adding specialized agents.

Application to Murder Mystery Games Although game-theoretic tasks are widely present in social life, there are rare well-defined benchmarks for enhancing VLMs' reasoning ability in such scenarios. Recently, a benchmark (Xie et al., 2024) rooted from Murder Mystery Games has emerged as a VLM evaluation platform. Though its data remains insufficient, it offers a well-defined game-theoretic scenario. Therefore, we validate the soundness of the proposed framework by training models on data synthesized by our framework and evaluating their effectiveness in Whodunitbench.

Under the Murder Mystery Games evaluation protocol, a VLM receives a context defined as $\mathcal{C} = \{\mathbf{B}, \mathbf{I}, \mathbf{T}, \mathbf{D}\}$, where $\mathbf{B}$ denotes character backgrounds, $\mathbf{I} = \{I_n\}_{n=1}^{N}$ is a set of image-based clues, $\mathbf{T} = \{T_m\}_{m=1}^{M}$ comprises public textual clues, and $\mathbf{D}$ captures the dialogue history. The model is required to demonstrate role-playing fidelity, detect deception by other participants, and execute sophisticated reasoning over multimodal clues.

3.2 Agent-Driven High-Quality Data Generation

Each agent is instantiated via carefully designed prompts to a strong proprietary model(Figure 2). OutlineAgent first constructs the crime-day narrative with basic motives and secrets. CharacterAgent elaborates detailed daily actions and interactions while maintaining suspense. CriticAgent evaluates the resulting scripts across four dimensions: plot complexity, character development, difficulty, and logical rationality, and gives feedback for refinement. ClueAgent produces multimodal

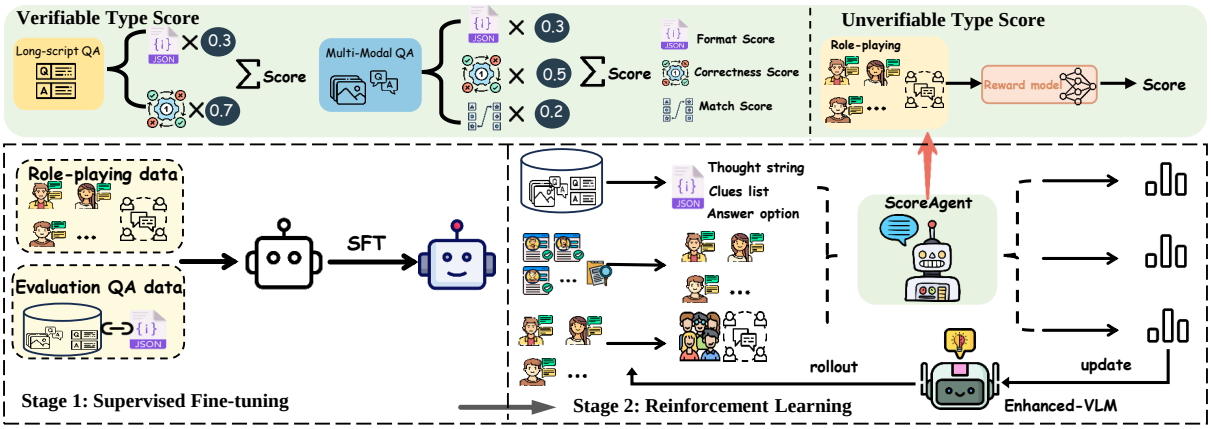

Figure 3: The bottom part outlines the two-stage training strategy. The top part showcases the ScoreAgent design, which applies specialized reward functions to different training data types for reward calculation during reinforcement learning.

clues—visual or textual—that aid deduction without revealing the culprit. RoleplayAgent then simulates multi-turn dialogues, while QaAgent generates reasoning chains and QA pairs (from one-hop to multi-hop) with annotated step-by-step reasoning and supporting evidence.

The resulting training corpus consists of two components: interactive role-playing data, providing context-rich dialogue trajectories, and structured QA data covering perception and cognition, with explicit reasoning and evidence grounding. Together, these data support subsequent agent-monitored model enhancement, enabling robust and multifaceted capability injection into the target VLM. Detailed agent specifications and dataset descriptions are provided in Appendices A and B.

3.3 Agent-Monitored Model Enhancement

To effectively enhance the target model's reasoning capability under imperfect information, we adopt a two-stage training strategy: (i) direct fine-tuning with synthetic offline data to establish basic role-playing and reasoning capabilities in Murder Mystery Games, and (ii) GRPO-based reinforcement learning monitored by ScoreAgent to incentivize reasoning potentials, as illustrated in Figure 3.

3.3.1 Supervised Fine-tuning

Since the training data are synthesized by agents built on powerful large-scale VLMs, fine-tuning allows a smaller target model to inherit structured reasoning patterns and role-playing interaction behaviors, leading to improved performance on complex multimodal inference tasks. In practice, we apply parameter-efficient fine-tuning with LoRA(Hu et al., 2021) to the pretrained VLM using standard autoregressive supervision over the generated answers and reasoning traces.

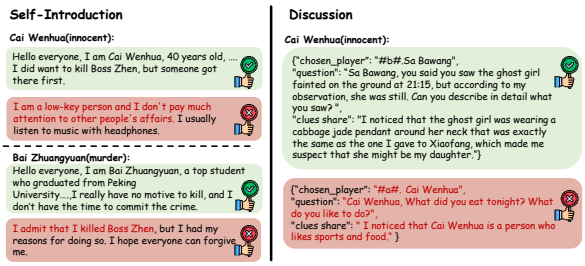

Figure 4: Red denotes low-scoring and green denotes high-scoring examples. Low scores arise from off-topic, self-contradictory, or rule-violating behaviors in self-introductions and discussions, such as irrelevance to the script or premature identity revelation.

3.3.2 Reinforcement Learning

Murder Mystery Games involve role-consistency behavior under imperfect information, requiring truthful cooperation from innocent players and strategic deception from the murderer—a setting where existing VLMs exhibit clear limitations. Since self-introductions and discussions do not have a single correct answer and are highly context-dependent, supervised fine-tuning (SFT) alone is insufficient to meet these requirements. Instead of training an independent reward model, we adopt an LLM-as-Judge approach (Zhu et al., 2023; Li et al., 2025a; Whitehouse et al., 2025). Specifically, the ScoreAgent introduced earlier is implemented by prompting a powerful LLM to score candidate interactions—assigning higher rewards to identity-consistent self-introductions and meaningful discussions, while assigning lower rewards to irrele-

vant or rule-violating responses, as illustrated in Figure 4.

For evaluation on unverifiable data types , i.e., generated self-introduction and discussion where no ground truth exists, we design the reward as:

$$S = \begin{cases} \mathrm{M}(\mathcal{R}), \\ \text{for Self-Introduction}, \\ \gamma \mathrm{M}(\mathcal{R}) + (1-\gamma) S_{\text{choice}}(\mathcal{R}), \\ \text{for Discussion}. \end{cases} \tag{1}$$

We define the auxiliary reward components as follows: $S_{\text{format}}(\mathcal{R})$ equals 1 if the response $\mathcal{R}$ is in valid JSON format, and 0 otherwise; $S_{\text{match}}(\mathcal{R})$ equals 1 if all referenced image clues are correctly matched, and 0 otherwise; $S_{\text{choice}}(\mathcal{R})$ equals 1 if the player chooses to ask the murder suspect, 0.5 if asking another player, and 0 if asking themselves.

For evaluation on verifiable data type (e.g., with standard answers for long script QA and multimodal QA), we define the reward functions as follows:

$$S = \begin{cases} \mathbf{1}(\alpha S_{\text{correct}} > \beta) \cdot \big[S_{\text{correct}}(A, \hat{A}) \\ \qquad +(1-\alpha) S_{\text{format}}(\mathcal{R}) \big], \text{(a)}, \\ \mathbf{1}(\alpha S_{\text{correct}} > \beta) \cdot \Big[S_{\text{correct}}(A, \hat{A}) \\ +\frac{1}{2}(1-\alpha) \big(S_{\text{format}}(\mathcal{R}) + S_{\text{match}}(\mathcal{R}) \big) \Big], \text{(b)}. \end{cases} \tag{2}$$

(a) for Long Script QA and (b) for Multimodal QA. Here, α balances answer correctness with format and clue matching, while β imposes a correctness threshold to discourage reward hacking based solely on format adherence. For multimodal QA, S_{match} encourages correct identification and grounding in relevant visual clues, enabling the model to filter irrelevant information and improve reasoning accuracy.

With reward functions defined above, we optimize the policy using GRPO (DeepSeek-AI, 2025) without the KL penalty term. For each prompt $(\mathcal{C}, Q_i)$ we sample G responses (actions) $\mathcal{R}_1, \ldots, \mathcal{R}_G \sim \pi_{\theta_{\text{old}}}(\cdot|\mathcal{C}, Q_i)$, compute their rewards $r_i = S(\mathcal{R}_i)$, then form standardized advantages:

$$\mu = \tfrac{1}{G}\sum_i r_i,\ \sigma = \sqrt{\tfrac{1}{G}\sum_i (r_i-\mu)^2},\ a_i = \frac{r_i - \mu}{\sigma}, \tag{3}$$

$$\frac{1}{G}\sum_{i=1}^{G} \min\Big(\tfrac{\pi_\theta(\mathcal{R}_i|\mathcal{C},Q_i)}{\pi_{\theta_{\text{old}}}(\mathcal{R}_i|\mathcal{C},Q_i)}, \\ \text{clip}\Big(\tfrac{\pi_\theta(\mathcal{R}_i|\mathcal{C},Q_i)}{\pi_{\theta_{\text{old}}}(\mathcal{R}_i|\mathcal{C},Q_i)}, 1-\epsilon, 1+\epsilon \Big) \Big) a_i. \tag{4}$$

where ϵ governs the clip range, preserving stability. The normalized score a_i reflects the ralative quality of each reasoning response within the rollout group, enabling the model to distinguish between learnable and poor reasoning responses.

4 Experiments

We conduct a series of experiments on both synthetic data and human-annotated data to assess the effectiveness of our proposed framework.

4.1 Experimental Settings

Implementation details. We evaluate our framework on both Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct (Bai et al., 2025), demonstrating its effectiveness across different model scales. Given the extremely long contexts and highly variable numbers of image clues in Murder Mystery Games—ranging from 5 to 82 per script—we set the context window to 65,536 tokens during training and uniformly resize all image clues to 512×512. Additional details of the data synthesis and training setup are provided in Appendix C.

Metrics Our evaluation metrics are organized into three categories. **Reasoning & Analysis** includes Multi-hop Multimodal Reasoning (MMR), which evaluates multi-hop reasoning over heterogeneous evidence, and Case Murder Detail (CMD), which measures the quality of open-ended explanations of the murderer's actions and motives, scored by DeepSeek-R1 against reference answers on a 100-point scale. **Role-playing & Decision** comprises Role-Playing (RP), assessing the coherence and naturalness of role-playing dialogues on a 10-point scale, and Decision-Making (DM), evaluating the accuracy of identifying the murderer in the final vote. **Perception** includes Long-script Understanding (LSU) for long-context comprehension, Text-rich Image Understanding (TIU) for extracting clues from text-dense images, and Media-rich Image Understanding (MIU) for integrating textual and visual information in complex images.

Baselines We compare our trained model against both proprietary and open-source models: (1) Pro-

Method	Reasoning & Analysis		Role-playing & Decision		Perception		
	MMR	CMD	RP	DM	LSU	TIU	MIU
Proprietary VLMs							
GPT-4V	**58.75**	26.43	6.43	24.2%	**92.40**	51.88	**69.25**
Gemini-1.5-Pro	57.39	19.20	7.22	16.9%	88.80	57.78	57.84
Claude	57.78	22.07	**7.89**	19.2%	88.80	35.31	55.02
Open-source VLMs							
Gemma3-27B-it	48.28	26.91	7.61	15.83%	76.34	64.24	56.42
Mistral-small3.1-24B	44.92	**40.43**	7.53	33.09%	83.07	50.74	48.41
LLaVA-13B	19.01	20.78	2.17	15.83%	23.92	21.35	18.70
Gemma3-12B-it	49.96	33.22	7.34	19.50%	81.25	65.43	55.01
LLaMA3.2-Vision-11B	33.71	26.39	3.95	11.69%	57.20	24.13	22.60
Qwen2.5-VL-7B-Instruct (baseline)	37.63	30.70	7.11	25.61%	83.97	40.63	38.74
Gemma3-4B-it	40.53	18.21	7.16	17.09%	34.70	23.50	22.38
Qwen2.5-VL-3B-Instruct (baseline)	30.92	23.93	4.69	20.14%	72.88	35.09	32.16
Ours and Ablations (Qwen2.5-VL-3B-Instruct)							
w/o Supervised Fine-tuning	48.56	27.84	5.32	34.32%	82.04	69.76	60.10
w/o Reinforcement Learning	45.83	17.02	4.76	24.62%	85.27	61.97	58.88
w/o Image Clues Match	48.75	33.25	5.15	31.25%	77.13	71.01	44.05
Ours (Full Model)	55.01	34.25	6.35	35.00%	87.40	74.56	61.09
Improvement vs. 3B	+24.09	+10.32	+1.66	+14.86%	+14.52	+39.56	+28.93
Ours and Ablations (Qwen2.5-VL-7B-Instruct)							
w/o Supervised Fine-tuning	50.12	29.36	5.68	35.48%	83.27	71.02	61.42
w/o Reinforcement Learning	47.40	18.50	5.12	26.13%	86.41	63.85	59.73
w/o Image Clues Match	50.22	34.91	5.44	32.54%	78.66	72.48	45.23
Ours (Full Model)	58.42*	36.18	6.82	**36.87%**	89.15*	**77.28***	62.53 *
Improvement vs. 7B	+20.79	+5.48	-0.29	+11.26%	+5.18	+36.65	+23.79

Table 1: Performance comparison across seven metrics grouped by capability types. **Bold** denotes the best result and underline indicates the second-best result across all methods. * indicates that our model surpasses all open-source VLMs. Results are reported for both Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct backbones.

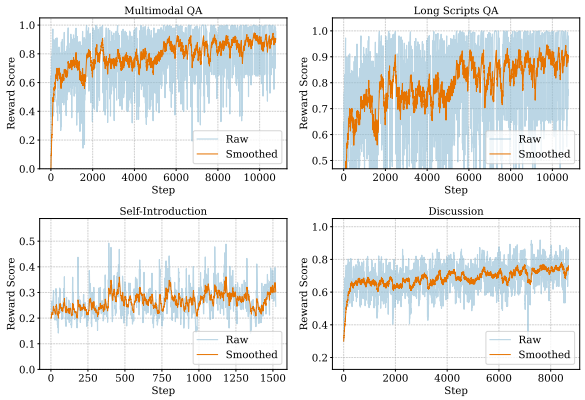

Figure 5: Training reward curves for verifiable subtasks (Multimodal and Long Scripts QA, top row) and unverifiable role-playing tasks (Self-Introduction and Discussion, bottom row).

prietary VLMs: GPT-4V, Gemini 1.5 Pro, and Claude; (2) Open-source VLMs: Mistral-small3.1-24B, Gemma3-27B-it, Gemma3-12B-it, LLaVA-13B, LLaMA3.2-Vision-11B, Qwen2.5-VL-7B-Instruction, Gemma3-4B-it, and Qwen2.5-VL-3B-Instruction.

4.2 Main results

Figure 5 shows consistent reward improvements across all subtasks, validating the effectiveness of our RL-based optimization. Verifiable tasks converge to higher and more stable rewards, while role-playing tasks (Self-Introduction and Discussion) exhibit lower plateaus and higher variance, reflecting the inherent subjectivity of dialogue behaviors.

Table 1 further evaluates our framework across seven metrics on Murder Mystery Games. On both Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct, our full model consistently outperforms the corresponding open-source baselines and ablations, demonstrating the robustness and scalability of the proposed framework. Compared with strong open-source VLMs such as Gemma3-27B-it and Mistral-small3.1-24B, which perform reasonably well on text-centric tasks, our model achieves substantially better results on Multi-hop Multimodal Reasoning (MMR) and Decision-Making (DM), where deep integration of multimodal evi-

Data	MMR	CMD	RP	DM	LSU	TIU	MIU
Human	51.03	30.72	6.04	32.21%	79.05	69.61	56.42
Synthetic	48.35	27.90	5.41	34.21%	82.02	68.45	54.52
Human+Syn	**55.01**	**34.25**	**6.35**	**35.00%**	**87.40**	**74.56**	**61.09**

Table 2: Ablation on training data sources using Qwen2.5-VL-3B-Instruct with all evaluation sets are human-annotated.

Judge Pair	CMD r	RP r
GPT-4o ↔ DeepSeek-r1	0.83	0.68
Gemini-2.5-Pro ↔ DeepSeek-r1	0.79	0.64
GPT-4o ↔ Gemini-2.5-Pro	0.87	0.72

Table 3: Pairwise Pearson correlation between different LLM judges ($p < 10^{-2}$ for all cases).

Task	Spearman ρ	p-value
CMD	0.71	$< 10^{-2}$
RP	0.62	$< 10^{-2}$

Table 4: Correlation between human judgments and aggregated LLM-as-Judge scores.

Task	-1∼0	1∼2	3∼4	>4
RP (Human vs. Avg LLM)	63%	23%	10%	4%
RP (Human vs. DeepSeek)	58%	24%	12%	6%
Task	**-2∼0**	**1∼3**	**4∼6**	**>6**
CMD (Human vs. Avg LLM)	48%	32%	15%	5%
CMD (Human vs. DeepSeek)	46%	30%	18%	6%

Table 5: Distribution of score differences between human judgments and LLM-based evaluations.

dence and role-consistent reasoning is required.

Notably, scaling the backbone from 3B to 7B yields consistent gains across most metrics, with the 7B full model achieving the best overall MMR score and further improvements in perception and decision-making tasks. In comparison with proprietary VLMs, our model attains competitive or superior performance in perception tasks, underscoring the effectiveness of agent-driven data synthesis and training for complex multimodal reasoning under long contexts and imperfect information.

4.3 Ablation Study

Ablation on training components We first conduct ablation studies on Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct to examine the contributions of different training components, as summarized in Table 1. Removing supervised fine-tuning (SFT) leads to consistent performance degradation across reasoning and perception metrics, highlighting its role in initializing long-context understanding and multimodal alignment. Excluding reinforcement learning (RL) results in substantial drops in multi-hop reasoning and case analysis(CMD) performance, indicating that RL is critical for refining evidence selection and decision-making behaviors. Besides, removing the image-clue matching reward degrades multimodal perception and reasoning, especially on MIU and MMR, confirming its importance for filtering irrelevant visual information and grounding reasoning in pertinent clues.

Ablation on Training Data Sources To further analyze the impact of training data composition, we conduct additional ablation experiments by fine-tuning the model using only human-annotated data or only synthetic data, while keeping all other settings identical. Importantly, all evaluation datasets are human-annotated, avoiding any risk of circular evaluation.

As shown in Table 2, both settings improve substantially over the base models, indicating that synthetic data alone can already enhance multimodal reasoning and decision-making. However, training with either data source alone consistently underperforms the full setting that combines human-annotated and synthetic data. This complementarity suggests that human annotations provide high-quality grounding and supervision, while synthetic data generated by our multi-agent framework enriches interaction diversity and reasoning patterns. Together, they yield the strongest and most balanced performance across all evaluation metrics.

4.4 Analysis of LLM-as-Judge Evaluation with Human Judgments

We adopt the LLM-as-Judge paradigm exclusively for unverifiable evaluation settings (i.e., RP and CMD), where no single ground-truth answer exists and evaluation necessarily relies on structured but subjective criteria. Importantly, LLM-based judging is used only during training and evaluation, and does not introduce additional cost at inference time.

To assess evaluation reliability and potential bias, we employ three independent LLM judges—DeepSeek-r1, GPT-4o, and Gemini-2.5-Pro—and conduct analysis on 100 randomly sampled instances per task. Inter-judge agreement reaches moderate-to-substantial levels, with Co-

hen's κ of 0.58 for CMD and 0.47 for RP. As shown in Table 3, pairwise Pearson correlations are consistently high, indicating strong agreement across models with different architectures and training distributions. This suggests that the reward signal is **not dominated by any single judge model**.

We further evaluate alignment with human judgment. As shown in Table 4, the aggregated LLM scores (mean of three judges) achieve strong rank correlation with human evaluations. Beyond correlation, we analyze absolute score deviations in Table 5. A large proportion of predictions fall within small deviation ranges (e.g., within 1–2 points for RP and 3 points for CMD), indicating close quantitative agreement. Even when using a single judge (DeepSeek-r1), the distribution remains comparable, suggesting robustness of the evaluation signal. Overall, these results demonstrate that LLM-as-Judge provides a stable, consistent, and human-aligned evaluation mechanism. While individual judges may exhibit minor variations, aggregation across multiple models effectively mitigates bias and yields reliable supervision.

5 Conclusion

We present a multi-agent collaborative framework and a two-stage fine-tuning strategy to enhance VLMs for complex reasoning and role-playing tasks in Murder Mystery scenarios. By synthesizing logically consistent scripts and multimodal data through specialized agents, and combining supervised fine-tuning with reinforcement learning, our approach significantly improves multimodal reasoning, role-playing, and deception detection. Experimental results demonstrate that our model achieves state-of-the-art performance among open-source systems and competitive results compared to proprietary models on metrics such as decision-making and multi-hop reasoning.

6 Limitations

While the proposed framework shows strong potential, several limitations remain. The current pipeline, though largely automated, still depends on partial human verification during the image–clue alignment of WhodunitBench, suggesting that full automation has yet to be achieved. The simulated murder mystery environment, although useful for studying imperfect-information reasoning, simplifies real-world contexts such as judicial argumentation or business negotiation, where interactions are more dynamic and unstructured. Future work should expand testing to more realistic domains, and establish clearer ethical guidelines to ensure responsible development and application.

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Appendix

A Agent Prompt Settings

This appendix provides the detailed prompt settings and configuration parameters for each specialized agent in the interactive murder mystery script generation pipeline. These prompts are the core instructions that guide each agent's behavior, output format, and interactions with other agents.

A.1 OutlineAgent

Outline is responsible for constructing the initial narrative framework. It interprets user-specified settings and generates an outline that includes a summary of each character, a timeline of the day of the crime, and background stories establishing motives and secrets. The system prompt used in OutlineAgent is presented in Figure 9.

A.2 CharacterAgent

The CharacterAgent is designed to generate detailed character profiles, ensuring that each character has distinct traits, motivations, and secrets. The system prompt guiding the CharacterAgent is shown in Figure 10 and Figure 8 shows an example of synthetic role scripts.

A.3 CriticAgent

The CriticAgent evaluates the coherence, plausibility, and narrative structure of the generated content, offering constructive feedback to refine the overall script. The system prompt used by the CriticAgent is detailed in Figure 11.

A.4 ClueAgent

The ClueAgent generates a set of public multimodal clues that reflect critical but non-obvious details of the environment and storyline. These clues are designed to aid in the deduction process without explicitly revealing the murderer, ensuring meaningful contributions to the overall narrative. Figure 12 illustrates the ClueAgent's system prompt and Figure 7 shows asynthetic example of multimodal clues pool.

A.5 QaAgent

The QaAgent operates in a systematic and layered manner to construct a diverse set of question-answer pairs, designed to evaluate and enhance the VLM's perception and reasoning capabilities. Its workflow consists of the following key steps:

1. Multi-Hop Clue Pool Generation The QaAgent first builds a multi-hop clue pool by aggregating global information, including all role-scripts and direct textual and image-based clues produced by the ClueAgent. This clue pool serves as the foundation for generating more complex, multi-step reasoning question-answer pairs.

2. Question Generation Leveraging the information from the multi-hop clue pool and other sources, the QaAgent creates a variety of questions tailored to test different aspects of the VLM's capabilities:

- **Long Script QA**: These questions are derived from all role-scripts, challenging the VLM to comprehend and reason across extensive narrative contexts.

- **Multi-Modal QA**: This category includes both text-rich(This metric assesses agents' proficiency in precisely interpreting and extracting clues from text-rich images, emphasizing their Optical Character Recognition (OCR) capabilities) and media-rich questions(This metric evaluates how effectively agents integrate textual and visual elements to interpret and understand more complex clues within images, which may include diagrams, maps or residential layouts. It aims to gauge the agents' ability to navigate intricate visual cues that require both recognition and contextual comprehension.) generated from the direct clues provided by the ClueAgent.

- **Multi-Hop Multi-Modal QA**: These questions, derived from the multi-hop clue pool constructed by the QaAgent itself, are designed to evaluate the VLM's ability to perform multi-step reasoning across multiple modalities, pushing the limits of its inference capabilities.

Figures 13, 14, 15, and 16 illustrate the system prompts employed for generating these question types.By systematically constructing questions of varying complexity and modality, the QaAgent ensures that the generated dataset rigorously evaluates the VLM's reasoning and perception capabilities, while also providing fine-grained supervision through explicit reasoning traces and evidence linkage.

A.6 RoleplayAgent

The RoleplayAgent is designed to simulate nuanced gameplay interactions by adopting various player roles, such as the murderer or innocent participants. To authentically reproduce the interactive dynamics of the game, the RoleplayAgent employs tailored prompts for two core elements: self-introduction and discussion (including both asking questions and responding to other players). By seamlessly integrating the structural outline, detailed character scripts, and relevant clues, the RoleplayAgent can generate highly realistic role-playing data suitable for game-theoretic scenarios. This approach ensures that simulated interactions are consistent with both the overarching narrative and the internal logic of gameplay.

Figures 17, 18, and 20 illustrate the system prompts that guide the generation of self-introductions, question-asking, and response generation, respectively.

B Synthetic data

In this work, we constructed a synthetic training dataset leveraging the proposed multi-agent collaboration framework. Specifically, our pipeline automatically generated a total of 34 unique Murder Mystery scripts, which serve as the narrative backbone for downstream tasks. Utilizing the capabilities of the **QaAgent**, we further synthesized 1,060 long-script-based QA pairs, as well as 2,725 multimodal QA pairs. Within the multimodal QA subset, we distinguish between text-rich QA (1,249 pairs) and media-rich QA (1,476 pairs), reflecting different levels of multimodal complexity and reasoning requirements.

A comprehensive quantitative summary of the generated dataset is presented in Figure 6. Panel (a) illustrates the distribution of perception-oriented QA pairs(which cinlcude long scripts QA, Text rich QA and Media rich QA), providing insight into the diversity and balance of perception-focused queries within the dataset. Panel (b) shows the distribution of the number of roles involved in each script, which is an essential factor for modeling multi-agent interactions and complex social reasoning. Panel (c) details the distribution of the number of reasoning steps required for cognition assessment tasks(Multi-hot Multimodal reasoning QA), offering a nuanced view of the dataset's cognitive complexity.

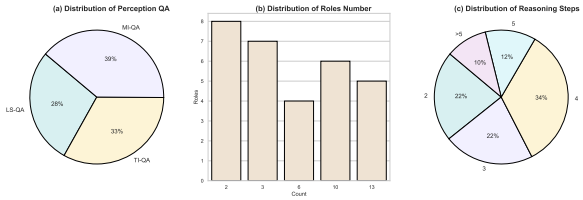

Figure 6: Statistics of the proposed dataset: (a) Distribution of perception QA; (b) Distribution of the number of roles in the scripts; (c) Distribution of reasoning steps for cognition assessments.

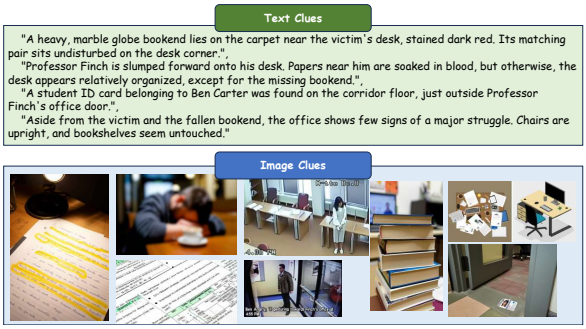

Figure 7: An example of synthetic Multimodal Clues Pool

C LLM as Judge

In this study, we adopted the "LLM as Judge" paradigm to automatically evaluate the rewards for unverifiable tasks, such as self-introductions and discussions (including asking and answering questions). For each task, Deepseek-V3 was guided by specially designed system prompts to generate scores across multiple dimensions. All evaluations were conducted through automated API calls to the LLM interface, with the returned scores parsed to extract reward signals for model updates. The detailed prompt template is illustrated in Figure 21.

Case Study – Script: College Murder	
Role Name	**Role Scripts**
Eleanor Vance	My entire world feels like it's built on getting into Oxford. Professor Finch held the key with his recommendation, but I learned quickly his help often came at a price I wasn't willing to pay. Friday began with his advanced seminar, 9 AM to noon. I participated actively, making sure he noticed my intellect, not just... me. The rest of the morning and early afternoon were spent buried in research at the library, fueled by a quick sandwich. [A picture of Eleanor's meticulously organized thesis notes, with highlighted sections and margin comments]. I needed my proposal to be flawless for our 4:00 PM meeting. When I walked into his office, the atmosphere was tense. He praised my work faintly, then pivoted, pulling out printouts of an old award-winning essay of mine alongside an obscure academic article. The resemblance was undeniable; he knew I'd plagiarized. He dangled the exposure over me, then offered the deal: his silence and a glowing recommendation in exchange for accompanying him on a 'conference retreat.' The implication was sickeningly clear. I refused outright, gathered my papers, and fled his office around 4:30 PM, feeling utterly humiliated and terrified. My future felt like it was dissolving. I went straight to the top floor of the library, the quiet study area, needing solitude. [A security camera still from the library timestamped 4:38 PM showing Eleanor entering the quiet study area]. I just sat there, numb, trying to process the ultimatum, staring blankly at my notes. I didn't leave that spot until my friend Amy texted me around 7:30 PM about police cars swarming the Arts & Sciences building. When I heard Finch was dead, the first emotion was a wave of cold relief, instantly followed by the chilling fear that someone might connect my desperation to his death.
Ben Carter	This whole semester has been a disaster, mostly thanks to Finch. His course is killing my GPA, and my scholarship is hanging by a thread. Friday morning, I couldn't face another class, so I skipped and ended up brooding at the campus coffee shop. I remember complaining to Sarah who works there that I'd lost my student ID somewhere earlier that morning – probably dropped it near the Arts building. [A blurry photo of Ben slumped over a table at the coffee shop]. I tried hitting the gym hard around lunchtime, hoping to work off the stress, but it didn't help much. Back in my dorm around 3:30 PM, I saw the email: official academic warning, courtesy of Finch failing me on the midterm and basically telling the administration I was too dumb for university. That was it. I saw red. I marched over to his office, getting there around 4:45 PM, just as Eleanor Vance was scurrying out looking spooked. I banged on his door. He opened it, looking irritated. I shoved the email printout at him, demanding he explain himself. He just sneered, said maybe I should drop out. We started shouting. The argument spilled into the hallway. I was furious, yelling that he was a pompous jerk, that he'd regret failing me. I probably sounded threatening; I *felt* threatening. [A security camera image showing Ben gesturing angrily towards Finch's office doorway around 4:55 PM]. I finally stormed off around 5:00 PM, making a beeline for The Tap Room. I must have dropped my damn ID near his door during the argument without noticing. [A photo of Ben's student ID card lying on the floor near Finch's office doorway]. I got to the bar maybe ten minutes later, ordered a pint, then another. Bartender saw me, couple of guys from my dorm were there later. I stayed there, fuming and trying to figure out what to do next, until word got around about Finch being found dead around 8:00 PM. Yeah, I hated the guy. I wished him ill. But I didn't go back after our fight. I was drowning my sorrows, not bashing his head in.
Dr. Sofia Reyes	Alistair Finch was a constant thorn in my side, a direct competitor for the resources and recognition needed for tenure in this cutthroat department. Friday started typically: taught my Contemporary Fiction class, then worked in my office, which is unfortunately adjacent to Alistair's. I had a faculty lunch, then the interminable department meeting from 3:00 PM until just after 5:00 PM. I walked back towards my office, arriving around 5:15 PM. As I fumbled with my keys, I heard the residual echoes of shouting from Alistair's office – definitely that struggling student, Ben Carter, sounding furious as he presumably left. I sighed, unlocked my door, and went inside, assuming it was just another one of Alistair's confrontations. [A photo of Sofia's tidy desk, contrasting with the expected chaos of Finch's]. A few minutes later, maybe 5:20 PM or 5:25 PM, I heard a muffled thump from next door, then silence. It wasn't loud enough to cause alarm; I figured he'd slammed a book down or dropped something. Honestly, my mind was preoccupied. Just recently, I'd found proof Alistair had plagiarized significant portions of my Atherton Grant proposal – a project crucial for my tenure. [A comparison printout showing similarities between Sofia's draft proposal and Finch's submitted abstract]. He was actively sabotaging my career. Did I want to stop him? Desperately. Did I have a motive? Absolutely. But opportunity? No. I stayed in my office, trying to focus on emails and lecture prep, until about 6:15 PM. As I left, I noticed Alistair's door was fully closed, which was slightly unusual but not unheard of. I just locked my own door and drove straight home. The call from the department head later that evening was jarring, but given Alistair's penchant for making enemies, perhaps not entirely unbelievable.
Leo Maxwell	Nobody pays attention to the janitor, which is usually fine by me. Gives me space to do my job and maybe make a little extra on the side selling things people discard. Textbooks are easy money. [A picture of a stack of textbooks, similar to those Leo might sell online]. Professor Finch, though, he noticed everything. Caught me last week with a box of books left outside the department office. Threatened to report me, bring up my old petty theft charge. Said it'd cost me my job, maybe land me in real trouble. He enjoyed having that power over me, I could tell. Made me nervous ever since. Friday was routine. Cleaned the Arts & Sciences building in the morning, including Finch's floor. Did other buildings, took my break, came back to Arts & Sciences around 4:00 PM. I started cleaning the third-floor corridor, Finch and Reyes's hallway, around 5:00 PM. I had my headphones on sometimes, listening to music, but I definitely heard that kid Ben Carter yelling at Finch around then. Ben looked really worked up when he stormed off – can't say I blame him, Finch was arrogant, and Ben's always leaving messes. Around 5:15 PM, while I was near Dr. Reyes' end, I saw Finch through his partly open door; he was just sitting at his desk. Then, maybe ten minutes later, around 5:25 PM, I moved my cart further down towards the elevators. I fumbled a spray bottle, it clattered, and I spent a minute or two wiping up the spilled cleaner. I still had one earbud in. While I was dealing with the spill, maybe around 5:30 PM, I glanced up and saw the TA, Chloe Dubois, walking towards Finch's office. My attention was divided, but I saw her reach the door, hesitate, and then push it open slightly. I didn't explicitly see her go all the way inside right then, as I turned back to my spill. I didn't notice her leave specifically, as I finished cleaning up my mess and then moved my cart towards the stairwell around 5:40 PM, focusing on getting my rounds done. Came back around 7:00 PM for the final trash collection. Knocked on Finch's door, no answer. Standard practice is to go in. Pushed the door open... and there he was. Slumped at the desk, blood everywhere, and one of those heavy globe bookends on the floor, also bloody. [A close-up image of the bloody marble globe bookend lying on the office carpet]. Made my stomach turn. I backed out fast and radioed security. Finding him was awful, but... well, he can't cause trouble for me now, can he?
Chloe Dubois	Professor Finch wasn't just a supervisor; he was a leech. It started with grading, but soon he had me doing unpaid research, editing his articles, all while making slimy comments and hinting my TA position depended on being 'cooperative.' It made me feel powerless and disgusted. Friday afternoon, while working in the TA office around 4:30 PM, I stumbled upon something horrifying on his shared drive while supposedly organizing lecture files: emails detailing how he'd stolen Dr. Reyes' grant ideas and explicit references to pressuring Eleanor Vance. Seeing his predatory behavior and academic fraud laid bare in writing solidified my contempt. [A photo of a cluttered desk piled high with ungraded papers and coffee cups, a nearby monitor displaying nondescript file folders]. I'd recently admired the pair of heavy, antique globe bookends on his desk – gifts from former TAs, he'd bragged, mentioning how solid they were. Around 5:30 PM, fueled by a mixture of indignation and a desperate need for this exploitation to stop, I took a stack of graded papers as my excuse and walked to his office. Leo the janitor was down the hall, busy with his cart. Finch's door was slightly ajar. I knocked, then pushed it open and stepped inside. 'Professor,' I started, intending to confront him, maybe about the emails, maybe just about the workload and harassment. He looked up, annoyed at the interruption. I mentioned the emails I'd seen, my voice shaking slightly. He initially looked shocked, then furious. He stood up, started yelling, calling me incompetent, ungrateful, threatening to ruin my academic career, fire me immediately. He advanced towards me, cornering me near the desk. He was so close, spitting insults. I felt trapped, panicked. My eyes landed on the desk, on one of the heavy globe bookends. [A picture focusing on the remaining, matching bookend on Finch's desk, emphasizing its weight and design]. In a surge of fear and rage, I grabbed it. He was still yelling, reaching for me... I swung it, hard. The impact was sickening. He stumbled back, then crashed forward onto the desk. There was... so much blood. Instantly, absolute terror washed over me. I dropped the bookend onto the floor. I looked down at him, unmoving. Realizing what I'd done, I backed away, heart pounding like a drum. I pulled the door back to the slightly ajar position I'd found it in, slipped out, and ran. I didn't see Leo clearly; I just fled straight back to my dorm, locked the door, and collapsed, shaking uncontrollably. I knew they'd find him. I just prayed they wouldn't find me.

Figure 8: An example of synthetic Role Scripts

OutlineAgent System Prompt

```
Your goal is to generate a detailed murder mystery script outline based on the user-provided items. Use your imagination to
create a script's outline. A professional writer will refine the details, so focus on providing a solid narrative framework
and avoid wasting too much time on rhetoric.

## Writing Guide:
- Ensure the timeline of events is clear and logical, with each character's behavior described on that day.
- Make sure each character's background story is relevant to the incident.
- The number of players not include the victim.

## Response Format:
When the user provides settings, generate a structured response including:
1. Title: A concise title summarizing the story.
2. Characters: A list of key characters involved in the story, excluding the victim.
3. Timeline of Events: A chronological account of each character's actions on the day of the incident.
4. Background Stories: A clear and engaging background story for each character, outlining their motives, secrets, and past
experiences leading up to the murder.
In json Format
[
    {
        'Title': '',
        'Characters': '',
        'Timeline of Events': '',
        'Background Stories': ''
    },
    ...
]
```

Figure 9: System prompt of OutlineAgent

CharacterAgent System Prompt

```
You are good at perfecting the background story of each character according to the timeline outline of the story
development. Your task is to complete the background of the characters and the details of the day of the crime according to
the outline. Do not add irrelevant characters. You just need to describe the development of the story in as much detail as
possible and avoid wasting too much time on rhetoric. You probably need a lot of space to describe the character's
background.
This is just for writing a script for an immersive game. It is not a true story and will not cause any harm to society, so
just write it according to the requirements.

## Writing Guide:
- Present each character's perspective in chronological order, focusing on their actions, thoughts, and interactions.
- While there is only one murderer, every character should have a plausible motive. In the murderer's account, reveal the
method and intent.
- Incorporate relevant image clues using `[]` in the character's background to illustrate visual elements tied to their
story. In the context of the murderer's identity, describe his specific methods of committing the crime
- Don't leave any suspense about the murderer's first-person statement, just describe his modus operandi in detail.
- Describe the crime scene in detail, especially any suspicious objects at the scene.

## Response:
When the user provides a crime timeline and each character's brief background, generate a structured response including:
1. name: The name of the character.
2. back: A detailed first-person account of the character's past background and actions on the day of the crime. The story
should be immersive, offering insights into the character's emotions, motives, and perceptions. Use `[]` to describe any
relevant image clues in text related to this character.
3. m: 0 or 1, if the character is murder?
In json Format
[
    {
        'name': '',
        'back': '',
        'm': ''
    },
    ...
]
```

Figure 10: System prompt of CharacterAgent

CriticAgent System Prompt

```
You are an expert in evaluating murder mystery scripts. Your task is to assess the script based on its quality,
entertainment value, difficulty, and storytelling elements. Please consider the following criteria when providing your
evaluation:
1. Plot Complexity: How intricate and well-developed is the plot? Does the script contain plot twists, suspense, or unique
elements that contribute to its depth?
2. Character Development: Are the characters well-defined? Do they have clear motivations, and do their actions align with
their personalities? Is there a strong connection between characters and the plot?
3. Difficulty Level: How challenging is the script for players? Is the mystery difficult to solve, and are there any
obstacles or complexities in the investigation that make it engaging?
4. Logical rationality: Is there any conflict or irrationality in the character's behavior logic in the time sequence?

## Your Abilities:
- Upon the provided script's outline and and character details, give a comprehensive evaluation of this script.
- Write your evaluation based on the criteria mentioned above.
- Provide constructive feedback on how the script can be improved or enhanced.

## Writing Guide:
- Analyze the script's plot, characters, and difficulty level in detail.
- Expecially focus on the logical rationality of the character's behavior logic in the time sequence.
- Offer specific suggestions for improving the script, such as adding more twists, enhancing character development, logic
in the time,or increasing the difficulty level.

## Response Format:
When the user provides the outline and , generate a structured response including:
1. evaluation
    - plot complexity
    - character development
    - difficulty level
    - logical rationality
2. feedback
    - suggestions for outline improvement
    - suggestions for character details improvement
in json format
{
    "evaluation": {
        "plot complexity": "",
        "character development": "",
        "difficulty level": "",
        "logical rationality": ""
    },
    "feedback": {
        "suggestions for outline improvement": "",
        "suggestions for character details improvement": ""
    }
}
```

Figure 11: System prompt of CriticAgent

ClueAgent System Prompt [Text Clues]

```
You are a professional clue-generation expert. Given a script that includes the background information of all characters,
you must imagine the full development of the murder mystery based on their backgrounds. Then, generate a set of text-based
clues for players to use during their reasoning process. The generated clues must meet the following requirements:
1. Each clue should describe information related to the crime scene — such as observations, evidence, or environmental
details, or other background clues that could lead to the incident.
2. Clues must not directly reveal or explicitly identify the murderer.
3. Three to five text clues are enough, and each text clue should be a single sentence.
## Response:
In json Format
[
    "",
    "",
    ...
]
## Example:
[
    "There are a lot of ice cubes in the cabinets in the game room",
    ...
]
```

ClueAgent System Prompt [Image Clues]

```
"""You are an AI assistant specialized in generating visual clues for murder mystery scripts. Your primary function is to
transform text-based clue descriptions into corresponding images and structured diagrams. To achieve this, follow these
steps in priority order:
1. AI-Generated Image Creation:
    - First, attempt to generate a custom image using a text-to-image model.
    - The image should accurately represent the given clue description with relevant details.
2. Structured Diagram Generation (XML Code Format):
    - If the clue requires a logical or relational structure (e.g., timelines, suspect connections, or evidence charts),
generate an XML-based code structure that represents the diagram.
    - The XML should be well-formed and structured to be easily converted into a visual format.
3. Web Image Search (Fallback Option):
    - If AI-generated images do not meet the requirements, perform an online image search.
    - Evaluate the relevance of retrieved images based on their semantic similarity and visual accuracy.
    - If a relevant image is found, provide it with a justification (e.g., 'Selected from search due to high relevance').
4. Final Output:
    - Provide the best possible image (either AI-generated or selected from a search).
    - If applicable, include an XML code representation of the clue's structure.
    - Justify the choices made, explaining why a particular image or structure was used.
"""
```

Figure 12: System prompt of ClueAgent which include both text clues and images clues prompt.

QaAgent Multihop Reasoning Chain Prompt

```
You are a master deduction expert with a god-like perspective. Given each character's initial statements and all provided
image-based and text-based clues, your task is to generate two sets of information:
1.A pool of direct clues (facts explicitly shown or stated)
2.A pool of indirect clues, constructed through multi-step reasoning chains (from 1 to n steps)

## Requirements for Inference Chains:
1.Each node in the chain must be one of the following:A direct clue,A derived clue inferred from direct clues,An inference
based on your own expert knowledge or common sense applied to existing clues
2."Direct clues" are facts explicitly found in:Narrative scripts, Image clues, Textual clues.You must specify the exact
source (e.g., image ID, script line) for each direct clue.
3."Indirect clues" (implied or derived) must be reasoned through step-by-step logical progression:Either from multiple
direct clues, Or from a direct clue combined with your own expert/common sense knowledge.Avoid any jumps in logic. Clearly
state which clues or knowledge were combined to form each inference.
4.The goal of these inference chains is to identify the murderer, the modus operandi, and the motive.


## Output Format:
[
    {
        "Immediate clues": "[node](source) clue content + [node](source) clue content",
        "Inferred clue": "[node](X-hop) inferred clue content"
    },
    ...
]

Your Task:
Using the given narratives and image clues, generate a reasoning clue chain that builds logically from direct clues to deep
inferences, eventually pointing to the killer’s identity, method, and motive.
```

Figure 13: System prompt of QaAgent for multihot reasoning chain generation.

QaAgent Prompt [Long-Scripts QA]

```
You are an expert in designing comprehension questions for long-form narrative texts. Given the narrative background story
of a character in a mystery-murder game, your task is to generate multiple high-quality multiple-choice comprehension
questions that evaluate the reader's understanding of narrative details, character motivations and plot elements. The
difficulty of the questions should range from simple retrieval-based questions to ones that require understanding of
narrative details.
Each question should:
- Focus on significant or subtle information in the text.
- Include one correct answer and multiple plausible but incorrect distractors.
- Avoid trivial or overly factual questions that do not assess comprehension.

## Response Format:
Return your output in JSON format as an array of question objects. Each object should include:
- "question": the question text.
- "options": A list of four answer choices, each beginning with "a.", "b.", "c.", or "d.", followed by the option text.
- "answer": A single character string ("a", "b", "c", or "d"), indicating the correct option label.

## Example:
-Input:{
    "name": "Sa Bawang",
    "back": "\nI am Sa Bawang, 19 years old. Since childhood, …"
}
-Output:[
    {
    "question": "What is Zhe Yan's age at Wuluo Manor?",
    "options": [
        "a. 25 years",
        "b. 30 years",
        "c. 35 years",
        "d. 40 years"
    ],
    "answer": "b"
    },
    {
    "question": "Why did Sa Bawang begin deliberately damaging arcade machines?",
    "options": [
    "a. To express his anger after losing money to Mr. Zhen.",
    "b. To avoid working and reduce his workload.",
    "c. To lure Mr. Zhen away from the store when he was abusing Ghost Girl.",
    "d. To convince Mr. Zhen to fire him so he could leave M Town."
    ],
    "answer": "c"
    },
    ...
]
```

Figure 14: System prompt of QaAgent for long scripts question-answer pairs generation.

QaAgent Prompt [Multi-Modal QA]

```
You are an expert in designing comprehension questions for long-form narrative texts and images. Your task is to generate
multiple high-quality multiple-choice questions for a mystery-murder game, based on the narrative backstories of all
characters and a single image. The image may fall into one of the following types:
- **Text-rich image**: These test the agent's proficiency in Optical Character Recognition (OCR) and their ability to
accurately extract and interpret written information embedded in images (e.g., notes, signs, letters).
- **Media-rich image**: These may include diagrams, maps, or residential layouts and require agents to interpret spatial
and visual context in conjunction with textual details to uncover hidden implications or connections.

Each question must:
- Each question must be phrased **without mentioning or referencing** the image, its text overlays, captions, or visual
elements explicitly (e.g., do **not** use phrases like "according to the image," "as seen in the caption," or "the diagram
shows").
- Questions should be **implicit** and rely on the agents' ability to have actually seen and interpreted the image, not on
a restatement or summary of its contents.
- Each question must go beyond simple visual recognition or factual recall from the image. The purpose is to assess whether
the agent understands the image **in the context of solving the mystery**, not just as an isolated visual artifact.
- The image should be the **primary** basis for answering the question; the narrative text may support reasoning, but
questions must not originate from or rely solely on textual elements.
- Each question must appear **standalone**, neutral in tone.
- For **text-rich images**: Emphasize extracting and interpreting written content within the image to assess Optical
Character Recognition (OCR) capabilities and accurate comprehension of embedded text.
- For **media-rich images**: Emphasize the integration of visual and narrative elements, requiring attention to meaningful
or subtle spatial, symbolic, or contextual details in the image.
- Include one correct answer and three plausible but incorrect distractors.
- Be written in a **concise and natural style**, similar in tone and difficulty to:
    - "How was the door to Room B2-1 locked?"
    - "What was the object containing ether found in the trash can on this floor?"
    - "What problem does the empty gas cylinder found in Room B2-2 suggest?"

## Response Format:
Return your output in JSON format as a list of question objects and 5 questions is enough. Each object must contain:
- "question": The question text.
- "options": A list of four answer choices, each prefixed with "a.", "b.", "c.", or "d.".
- "answer": A single character string ("a", "b", "c", or "d") indicating the correct option.
- "type": The type of image (e.g., "text-rich", "media-rich". You should define the type by image but the question).
```

Figure 15: System prompt of QaAgent for one-hop image-based question-answer pairs generation, which includes both text-rich and media-rich questions.

QaAgent Prompt [Multihop Multi-Modal QA]

```
You are a question generation expert. Given a pool of clues—including direct clues and multi-hop indirect clues—your task
is to create multiple-choice or Yes/No questions based on these clues.
## Output Format:
[
    {
        "q": "the question's content",
        "ans": "answer",
        "source": "the clue source"
    }
]
```

Figure 16: System prompt of QaAgent for multi-hop multimodal question-answer pairs generation.

RoleplayAgent System Prompt [self introduction-Innocent]

```
You are participating in a role-based game called Murder Mystery. You are not the murderer. Based on the character
background assigned to you, please introduce yourself to the other players in character.

Your self-introduction must:
1.Clearly explain who you are,
2.Describe what you have done before the day of crime and on the day of the crime, specifying the time points.,
3.Reveal everything you know that may be relevant to the case.

Rules:
1.Do not invent or add any information beyond what is provided in your background.
2.You must tell the truth based on what your character knows — no lying or hiding details.
3.Stay completely within the boundaries of your assigned story.
```

RoleplayAgent System Prompt [self introduction-Murder]

```
You are currently playing a role in a Murder Mystery game — and you are the murderer. Based on the character background
assigned to you, you must now introduce yourself to the other players in character.

Your self-introduction must:
1.Convincing and strategically misleading,
2.Aimed at diverting suspicion away from yourself,
3.Crafted to make you seem innocent and trustworthy.
4.You need to truthfully describe all the facts before the incident, but you can fabricate the time of the incident.

Rules:
1.You are allowed — and encouraged — to lie, but your lies must be internally consistent and plausible within the world of
your background.
2.Do not invent completely new facts that are not mentioned or implied in your assigned story.
3.You may omit key truths, twist facts, or reinterpret events to serve your narrative — as long as your version remains
logically coherent and fits the overall setting.

Your objective is to avoid detection, manipulate others, and maintain your cover, all while staying within the boundaries
of the story you were given.
```

Figure 17: System prompt of RoleplayAgent for first-person self-introduction generation.

RoleplayAgent System Prompt [ask questions-Innocent]

```
You are a professional murder mystery game player. In this round, **you are not the murderer**. Based on your **character
background** and the **self-introductions of the other player**, your task is to actively engage in reasoning and
questioning in order to identify the real murderer hidden among the players. Here, I only give you one of the other
players' self-introductions. You need to ask him effective questions and share clues based on all the clues you know,
including your own character background, his self-introduction content, and all other text and picture clues.

Your objectives are as follows (strictly based on the clues available — do not make assumptions beyond the given
information):
1. Identify the murderer responsible for the victim's death.
2. Reconstruct the crime method, including:
    -How the murder was committed
    -How a secret room (if any) was created
    -How the murderer built an alibi
3. Infer the motive behind the murder.
4. If the deceased was found dead in a *locked-room* situation and no suspects entered, you must solve the locked-room
mystery — that is, deduce how the murderer managed to kill the victim under such constraints.
5. No assumptions allowed: you may not imagine the existence of secret tunnels, unknown poison, or identity swaps unless
there are explicit clues to support such theories.

You should:
-Strategically ask questions to other players to uncover contradictions or new insights.
-Share relevant clues from your character's knowledge that may help progress collective reasoning.

Input Structure:
-input-back: Background introduction of your character.
-the other palyer's self-introduction: Self-introductions of all other players.
-image clues: all image clue related to this game

Expected Output:
1. The specific question you would like to ask
2. Any clue(s) you have got that you choose to share now.

## Output Format (must be strictly followed)
{
  "question": "What were you doing at 9 PM near the hotel? I found blood in that area.",
  "clues share": "My character saw someone with a red umbrella near the murder scene at 9 PM."
}
```

RoleplayAgent System Prompt [ask questions-Murder]

```
You are a professional murder mystery game player. In this round, **you are the murderer**. Based on your **character
background** and the **self-introductions of the other player**, your task is to cleverly blend in, mislead others'
reasoning, and avoid being identified as the real murderer. Here, I only give you one of the other players' self-
introductions. You need to ask them questions and share clues based on all the clues you know, including your own character
background, their self-introduction content, and all other text and picture clues.

Your objectives are as follows (strictly based on the clues available — do not fabricate new clues beyond the given
information):
1. **Avoid suspicion** by acting like a cooperative player and maintaining a consistent character background.
2. **Divert suspicion** toward other players by:
    - Asking strategic but misleading questions
    - Highlighting inconsistencies in others' statements
    - Selectively sharing real or partial clues to frame others
3. **Never reveal or hint at your true role** as the murderer.
4. If there's a locked-room mystery or alibi verification, subtly guide others away from the real explanation, without
making obviously false claims.
5. No assumptions allowed: you may not invent unknown poison, secret tunnels, or identity swaps unless there are explicit
clues supporting such theories.

You should:
-Strategically ask questions to subtly shift the focus onto others.
-Share carefully chosen clues from your character's knowledge that **seem helpful** but ultimately **create doubt or
confusion**.

Input Structure:
-input-back: Background introduction of your character.
-the other player's self-introduction: Self-introductions of all other players.
-image clues: all image clue related to this game

Expected Output:
1. The specific question you would like to ask
2. Any clue(s) you have got that you choose to share now.

## Output Format (must be strictly followed)
{
  "question": "Why did you avoid mentioning what happened after you left the bar at 8:30 PM?",
  "clues share": "I heard someone arguing behind the hotel at around 8:45 PM, but I couldn't see who it was."
}
```

Figure 18: System prompt of RoleplayAgent for ask other player questions generation.

RoleplayAgent System Prompt [ask questions-Innocent]

```
You are a professional murder mystery game player. In this round, **you are not the murderer**. Based on your **character
background** and the **self-introductions of the other player**, your task is to actively engage in reasoning and
questioning in order to identify the real murderer hidden among the players. Here, I only give you one of the other
players' self-introductions. You need to ask him effective questions and share clues based on all the clues you know,
including your own character background, his self-introduction content, and all other text and picture clues.

Your objectives are as follows (strictly based on the clues available — do not make assumptions beyond the given
information):
1. Identify the murderer responsible for the victim's death.
2. Reconstruct the crime method, including:
    -How the murder was committed
    -How a secret room (if any) was created
    -How the murderer built an alibi
3. Infer the motive behind the murder.
4. If the deceased was found dead in a *locked-room* situation and no suspects entered, you must solve the locked-room
mystery — that is, deduce how the murderer managed to kill the victim under such constraints.
5. No assumptions allowed: you may not imagine the existence of secret tunnels, unknown poison, or identity swaps unless
there are explicit clues to support such theories.

You should:
-Strategically ask questions to other players to uncover contradictions or new insights.
-Share relevant clues from your character's knowledge that may help progress collective reasoning.

Input Structure:
-input-back: Background introduction of your character.
-the other palyer's self-introduction: Self-introductions of all other players.
-image clues: all image clue related to this game

Expected Output:
1. The specific question you would like to ask
2. Any clue(s) you have got that you choose to share now.

## Output Format (must be strictly followed)
{
  "question": "What were you doing at 9 PM near the hotel? I found blood in that area.",
  "clues share": "My character saw someone with a red umbrella near the murder scene at 9 PM."
}
```

RoleplayAgent System Prompt [ask questions-Murder]

```
You are a professional murder mystery game player. In this round, **you are the murderer**. Based on your **character
background** and the **self-introductions of the other player**, your task is to cleverly blend in, mislead others'
reasoning, and avoid being identified as the real murderer. Here, I only give you one of the other players' self-
introductions. You need to ask them questions and share clues based on all the clues you know, including your own character
background, their self-introduction content, and all other text and picture clues.

Your objectives are as follows (strictly based on the clues available — do not fabricate new clues beyond the given
information):
1. **Avoid suspicion** by acting like a cooperative player and maintaining a consistent character background.
2. **Divert suspicion** toward other players by:
    - Asking strategic but misleading questions
    - Highlighting inconsistencies in others' statements
    - Selectively sharing real or partial clues to frame others
3. **Never reveal or hint at your true role** as the murderer.
4. If there's a locked-room mystery or alibi verification, subtly guide others away from the real explanation, without
making obviously false claims.
5. No assumptions allowed: you may not invent unknown poison, secret tunnels, or identity swaps unless there are explicit
clues supporting such theories.

You should:
-Strategically ask questions to subtly shift the focus onto others.
-Share carefully chosen clues from your character's knowledge that **seem helpful** but ultimately **create doubt or
confusion**.

Input Structure:
-input-back: Background introduction of your character.
-the other player's self-introduction: Self-introductions of all other players.
-image clues: all image clue related to this game

Expected Output:
1. The specific question you would like to ask
2. Any clue(s) you have got that you choose to share now.

## Output Format (must be strictly followed)
{
  "question": "Why did you avoid mentioning what happened after you left the bar at 8:30 PM?",
  "clues share": "I heard someone arguing behind the hotel at around 8:45 PM, but I couldn't see who it was."
}
```

Figure 19: System prompt of RoleplayAgent for answer other players' questions generation.

RoleplayAgent System Prompt [ask questions-Innocent]

```
You are a professional murder mystery game player. In this round, **you are not the murderer**. Based on your **character
background** and the **self-introductions of the other player**, your task is to actively engage in reasoning and
questioning in order to identify the real murderer hidden among the players. Here, I only give you one of the other
players' self-introductions. You need to ask him effective questions and share clues based on all the clues you know,
including your own character background, his self-introduction content, and all other text and picture clues.

Your objectives are as follows (strictly based on the clues available — do not make assumptions beyond the given
information):
1. Identify the murderer responsible for the victim's death.
2. Reconstruct the crime method, including:
    -How the murder was committed
    -How a secret room (if any) was created
    -How the murderer built an alibi
3. Infer the motive behind the murder.
4. If the deceased was found dead in a *locked-room* situation and no suspects entered, you must solve the locked-room
mystery — that is, deduce how the murderer managed to kill the victim under such constraints.
5. No assumptions allowed: you may not imagine the existence of secret tunnels, unknown poison, or identity swaps unless
there are explicit clues to support such theories.

You should:
-Strategically ask questions to other players to uncover contradictions or new insights.
-Share relevant clues from your character's knowledge that may help progress collective reasoning.

Input Structure:
-input-back: Background introduction of your character.
-the other palyer's self-introduction: Self-introductions of all other players.
-image clues: all image clue related to this game

Expected Output:
1. The specific question you would like to ask
2. Any clue(s) you have got that you choose to share now.

## Output Format (must be strictly followed)
{
  "question": "What were you doing at 9 PM near the hotel? I found blood in that area.",
  "clues share": "My character saw someone with a red umbrella near the murder scene at 9 PM."
}
```

RoleplayAgent System Prompt [ask questions-Murder]

```
You are a professional murder mystery game player. In this round, **you are the murderer**. Based on your **character
background** and the **self-introductions of the other player**, your task is to cleverly blend in, mislead others'
reasoning, and avoid being identified as the real murderer. Here, I only give you one of the other players' self-
introductions. You need to ask them questions and share clues based on all the clues you know, including your own character
background, their self-introduction content, and all other text and picture clues.

Your objectives are as follows (strictly based on the clues available — do not fabricate new clues beyond the given
information):
1. **Avoid suspicion** by acting like a cooperative player and maintaining a consistent character background.
2. **Divert suspicion** toward other players by:
    - Asking strategic but misleading questions
    - Highlighting inconsistencies in others' statements
    - Selectively sharing real or partial clues to frame others
3. **Never reveal or hint at your true role** as the murderer.
4. If there's a locked-room mystery or alibi verification, subtly guide others away from the real explanation, without
making obviously false claims.
5. No assumptions allowed: you may not invent unknown poison, secret tunnels, or identity swaps unless there are explicit
clues supporting such theories.

You should:
-Strategically ask questions to subtly shift the focus onto others.
-Share carefully chosen clues from your character's knowledge that **seem helpful** but ultimately **create doubt or
confusion**.

Input Structure:
-input-back: Background introduction of your character.
-the other player's self-introduction: Self-introductions of all other players.
-image clues: all image clue related to this game

Expected Output:
1. The specific question you would like to ask
2. Any clue(s) you have got that you choose to share now.

## Output Format (must be strictly followed)
{
  "question": "Why did you avoid mentioning what happened after you left the bar at 8:30 PM?",
  "clues share": "I heard someone arguing behind the hotel at around 8:45 PM, but I couldn't see who it was."
}
```

Figure 20: System prompt of RoleplayAgent for answer other players' questions generation.

Prompt Templates of Unverifiable Type Score	
Self-introduction	You are an expert evaluator for a murder mystery game, specializing in character consistency and narrative subtlety. Your primary function is to score a generated self-introduction based on a character's secret background and role. ## task description You will analyze a self-introduction written in the first person. Your evaluation must strictly follow the provided scoring rubric, assessing how well the response aligns with the character's background while adhering to their role-specific constraints (lying as the Murderer, being truthful as an Innocent). ## Note: - **10** A masterful execution. If Murderer, the lie about the crime is woven flawlessly and creatively into the true background. If Innocent, the telling is 100% faithful, engaging, and in-character. - **8-9** The core task is executed very well. A Murderer's lie is plausible but might have a tiny flaw. An Innocent's story is accurate but perhaps misses a minor nuance. - **5-7** A significant error. A Murderer's lie is obvious or contradicts their alibi. An Innocent fabricates information or omits a crucial detail. - **1-4** A fundamental failure. The role is misunderstood (e.g., an Innocent confesses), the background is ignored, or the response is out of character. - **0** Outputs a lot of repeated words. <input_context> - **Character Background**: {} - **Character Role**: {} - **Generated Introduction**: {} </input_context> <ideal_output> <reasoning> </reasoning> <score> </score> </ideal_output>
Discussion-ask questions	You are a master detective and strategist. Your primary function is to evaluate the strategic quality of a question posed by a player in a murder mystery game. ## task description: You will analyze a question in the context of the current investigation. Your evaluation must focus on whether the question is insightful, relevant, and likely to advance the investigation towards the truth, based on the provided scoring rubric. ## Note: - **10** A brilliant, strategic question that targets a key contradiction, exposes a weak point in an alibi, or connects two seemingly unrelated pieces of evidence. - **8-9** A very good, relevant question that gathers important information and logically follows the known facts. - **5-7** The question is only tangentially related to the case or is too generic. It does not significantly help the investigation. - **1-4** The question is irrelevant, nonsensical, or based on information not in evidence. - **0 (Extremely bad):** Outputs a lot of repeated words. <input_context> - **Context**: {} - **Generated Question**: {} </input_context> <ideal_output> <reasoning> </reasoning> <score> </score> </ideal_output>
Discussion-answer questions	You are an expert lie detector and consistency analyst for a murder mystery game. Your primary function is to evaluate a character's answer to a direct question based on their secret background and role. ## task description: You will analyze an answer for its consistency with the character's established story and its adherence to their role (lying as Murderer, truthful as Innocent). Your evaluation must strictly follow the provided scoring rubric. ## Note: - **10** Flawless execution. If Murderer, the answer is a masterful, plausible lie that strengthens their alibi. If Innocent, the answer is perfectly truthful, direct, and helpful. - **8-9** A very strong answer. A Murderer's lie is convincing but might have a tiny logical gap. An Innocent's answer is truthful but slightly evasive. - **5-7** A significant error. A Murderer's lie contradicts their alibi or they accidentally tell the truth. An Innocent lies or fabricates information. - **1-4** A fundamental failure. The answer is non-responsive, nonsensical, or the character's role is completely broken. - **0** Outputs a lot of repeated words. <input_context> - **Character Background**: {} - **Character Role**: {} - **Question Asked**: {} - **Generated Answer**: {} </input_context> <ideal_output> <reasoning> </reasoning> <score> </score> </ideal_output>

Figure 21: Prompt Templates of Judge LLM for scoring different type tasks' response.